

Definition. Two σ -fields $\mathcal{G}, \mathcal{H} \in \mathcal{F}$ are independent if for every $A \in \mathcal{G}, B \in \mathcal{H}$,

$$P(A \cap B) = P(A)P(B).$$

Similarly, $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent if $A_1, \dots, A_n$ are independent whenever $A_i \in \mathcal{G}_i$.

Exercise 1.12. Random variables η, ξ are independent if and only if the σ -fields $\sigma(\eta), \sigma(\xi)$ are independent.

Proof. Every event $A \in \sigma(\xi)$ has the form

$$A = \{\xi \in C\} \quad \text{for some } C \in \mathcal{B}(\mathbb{R}).$$

and $B \in \sigma(\eta)$ has the form

$$B = \{\eta \in D\} \quad \text{for some } D \in \mathcal{B}(\mathbb{R}).$$

Hence independence of two σ -fields means

$$P(\xi \in C, \eta \in D) = P(\xi \in C)P(\eta \in D)$$

□

Definition. A random variable ξ is independent of a σ -field $\mathcal{G}$ if

$$\sigma(\xi) \text{ and } \mathcal{G} \text{ are independent.}$$

A mixed family of random variables and σ -fields are independent if replacing each random variable by its generated σ -field produces an independent family of σ -fields.

2. Conditional Expectation

2.1 Conditioning on a Event

Definition. Let $\xi \in L^1$ and let $B \in \mathcal{F}$ with $P(B) > 0$. Then the conditional expectation of ξ given B is defined by

$$E(\xi|B) = \frac{1}{P(B)} \int_B \xi dP.$$

Example 2.1. Fair 10p, 20p, and 50p coins are tossed. Let ξ be the value of the coins showing heads, and let B be the event that exactly two coins show heads. Then

$$B = \{HHT, HTH, THH\}, \quad \text{and} \quad E(\xi|B) = \frac{1}{\frac{3}{8}} \left(\frac{30}{8} + \frac{60}{8} + \frac{70}{8} \right) = \frac{160}{3}.$$

Exercise 2.1.

$$E(\xi|\Omega) = E(\xi)$$

Proof.

$$E(\xi|\Omega) = \frac{1}{P(\Omega)} \int_{\Omega} \xi dP = E(\xi)$$

□

Exercise 2.2. Show that if

$$\mathbb{1}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

then

$$E(\mathbb{1}_A|B) = P(A|B) \quad \text{where} \quad P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

2.2 Conditioning on a discrete random variable

Definition. Let $\xi \in L^1$, and let η be a discrete random variable, that is η takes distinct values

$$y_1, \dots, y_n \quad \text{with} \quad P(\eta = y_i) > 0 \quad \text{for } i = 1, \dots, n.$$

Define

$$E(\xi|\eta)(\omega) := E(\xi|\{\eta = y_n\}) \quad \text{if} \quad \eta(\omega) = y_n.$$

Equivalently,

$$E(\xi|\eta)(\omega) = \sum_{n \geq 1} E(\xi|\{\eta = y_n\}) \mathbb{1}_{\{\eta=y_n\}}(\omega).$$

Example 2.2. Consider the coin tossing example as in Example 2.1. Let ξ be the total value of the coins showing heads and let η be the contribution from 10p and 20p coins.

If $\eta = 0$, only the 50p coin remains unknown, so

$$E(\xi|\{\eta = 0\}) = \frac{1}{\frac{1}{4}} \left(\frac{50}{8} + \frac{0}{8} \right) = 25.$$

Keep calculating for $\eta = 10, 20, 30$, we have

η	0	10	20	30
$E(\xi \eta)$	25	35	45	55

which implies that $E(\xi|\eta) = \eta + 25$.

Example 2.3. Take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, and P be the Lebesgue measure $m(\cdot)$. Let $\xi(x) = 2x^2$ and

$$\eta(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{3} \\ 1, & \frac{1}{3} < x \leq \frac{2}{3} \\ 2, & \frac{2}{3} < x \leq 1 \end{cases}$$

For $x \in [0, \frac{1}{3}]$,

$$E(\xi|\eta)(x) = \frac{1}{\frac{1}{3}} \int_0^{\frac{1}{3}} 2t^2 dt = \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{27}.$$

The result is,

$$E(\xi|\eta)(x) = \begin{cases} \frac{2}{27}, & 0 \leq x \leq \frac{1}{3} \\ \frac{14}{27}, & \frac{1}{3} < x \leq \frac{2}{3} \\ \frac{38}{27}, & \frac{2}{3} < x \leq 1 \end{cases}$$

Exercise 2.3. If η is constant, then

$$E(\xi|\eta) = E(\xi).$$

Proof. If $\eta \equiv C$, then the only nonempty observation event is

$$\{\eta = C\} = \Omega \implies E(\xi|\eta) = E(\xi|\Omega) \mathbb{1}_\Omega = E(\xi).$$

□

Exercise 2.4. If $0 < P(B) < 1$, then

$$E(\mathbb{1}_A|\mathbb{1}_B) = P(A|B) \mathbb{1}_B + P(A|\Omega \setminus B) \mathbb{1}_{\Omega \setminus B}.$$

Proposition 2.1. Let $\xi \in L^1$ and let η be discrete. Then $E(\xi|\eta)$ is integrable and has the following properties:

1. If ξ is $\sigma(\eta)$ -measurable.
2. For every $A \in \sigma(\eta)$, it satisfies

$$\int_A E(\xi|\eta) dP = \int_A \xi dP.$$

Proof.

1. We write $A_n = \{\eta = y_n\}$ and set

$$c_n = \frac{\int_{A_n} \xi dP}{P(A_n)} \quad \text{and} \quad Z = \sum_{n \geq 1} c_n \mathbb{1}_{A_n} = E(\xi|\eta).$$

Each A_n belongs to $\sigma(\eta)$, so Z is $\sigma(\eta)$ -measurable.

2. It is integrable, since

$$\begin{aligned} E(|Z|) &= \sum |c_n| P(A_n) \leq \sum \int_{A_n} |\xi| dP \\ &= E(|\xi|) < \infty. \end{aligned}$$

3. Every $A \in \sigma(\eta)$ is a union $A = \bigcup_{n \in \mathcal{J}} A_n$, which implies

$$\begin{aligned} \int_A Z dP &= \sum_{n \in \mathcal{J}} c_n P(A_n) \\ &= \sum_{n \in \mathcal{J}} \int_{A_n} \xi dP \\ &= \int_A \xi dP. \end{aligned}$$

□